

RFP — Enshrined Liquid Staking on Flare (eFLR)

A design document for a protocol-owned liquid staking layer: what it would have to contain, which decisions it forces, and what would kill it.

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Requires (reading): FIP.02, FIP.05, FIP.10, FIP.16

Disclosure: the author holds FLR and sFLR, runs LP positions on Spectra with sFLR as underlying, has had occasional contact with the Spectra team, and has an interest in duration markets on Flare existing. Section 6 cuts against the sFLR position; Section 9 runs with the duration interest. Weigh accordingly.

"FIP.17" is used below as a placeholder. The next FIP number is the Foundation's to assign. "eFLR" (enshrined FLR) is likewise a working name.

1. Executive Summary

Flare limits staking concentration by counting boxes: four validator nodes per registered entity, 300 million FLR per node, a 15× delegation factor on a 1 million FLR self-bond. The limit is enforced by a C-chain registry, not by the P-chain, and it constrains the entity that runs validators. It cannot see the pool that delegates to them. The largest pool on the network holds 2.384 billion FLR, 11.08% of active stake, and is subject to no cap of any kind. The September 2026 Sceptre case showed what a rational operator does when the permitted route costs 20% of rewards and the alternative is a second registration: it registers.

A percentage cap per entity, floating with the network, is the first fix and belongs in FIP.02/FIP.05, not here. This document is about the second problem, the one a percentage cap does not touch. Capital that is capped on one route moves to another. A pool that may not run its own boxes delegates to strangers and buys them back with fee rebates. A proxy with kickbacks breaks no rule on the books and every intention behind them, and governance can chase it for years. The alternative is to stop regulating the pool and to own it: a protocol-issued liquid staking token, eFLR, whose delegation is routed by a published formula across the validator set, with a per-entity cap the chain enforces because the chain is the delegator. The router does not replace competition among validators; it moves the competition from access to pool operators to measured performance, and every entity competes for the same delegation on the same terms.

The precedent is FIP.16, within limits Section 2.4 draws: faced with MEV, Flare did not write a rule against extractors, it moved block building to a designated builder. Private liquid staking is extraction of validator selection and of custody over pooled capital, and the same move is available for the first of those; the second needs a settlement primitive Flare does not have and this RFP asks for as its own FIP.

Three scopes: an enshrined router that pools above a threshold must use, enforced as a reward-eligibility rule rather than a mandate (A); a protocol-issued token under protocol custody (B); both (C). The recommendation is A now and B only if custody can be shown in a key matrix, not promised in prose. The success metric for A and C is one number, the share of pooled stake that reaches validators through the router; eFLR's own market share is a metric for B and a distraction for the rest.

The document is a request for proposal, not a specification. Numbers are proposals with reasons; the ones the argument depends on are in Appendix B with dates. The failure modes in Section 11 start with the one that can kill the design: an enshrined pool is the largest pool on the network, and its neutrality is whether one party holds image, upgrade key, feeds and parameter initiative at once.

2. The Problem: Capacity Is Not Independence

2.1 What the current rules measure

FIP.05 (September 2023) set the staking parameters that still govern: minimum self-bond 1 million FLR, delegation factor 15×, maximum 200 million FLR per node, up to four validators per infrastructure entity, uptime above 80% to earn rewards. The FIP.16 hard fork of 14 July 2026 raised the node maximum to 300 million. The entity ceiling is therefore 1.2 billion FLR.

Three things about that ceiling. It is absolute on a network whose active stake went from roughly 11 billion in March 2026 to 21.5 billion by September; a fixed number in a doubling system is a limit on growth wearing the costume of a limit on concentration. It is enforced by the entity registry on the C-chain, not by the P-chain, which FIP.05 says in plain words. And it measures the wrong layer. A node is a validator with its own stake; an entity is the registration that may run four of them; a pool is a contract that delegates to nodes it does or does not own. The rule constrains entities. The concentration sits in pools.

The one percentage that does exist, the 5% vote-power cap per validator above which delegators are diluted rather than refused, is slack: under a 300 million node cap no single node reaches 1.075 billion.

2.2 What FIP.16 changed

FIP.16 (April 2026) did four things that matter here. It cut inflation from 5% to 3%. It set a mandatory minimum 20% entity fee on everything staked or delegated to an

entity. It raised the weight of P-chain stake to $5\times$ C-chain delegation in the signing weight that governs FTSO, FDC and block-latency rewards, which pushes capital from liquid WFLR delegation into locked P-chain stake. And it created FIRE, the Flare Income Reinvestment Entity, with a mandate to capture fees and MEV and a roadmap that moves block building to a FIRE-designated builder in three stages.

Put together: the network made third-party delegation more expensive (20% floor), made P-chain staking more rewarding ($5\times$ weight), and left the enforcement of the entity limit to a registry that counts addresses. A liquid staking pool above 1.2 billion FLR now pays 20% on everything above the ceiling unless it finds a way around the ceiling. Sceptre found one. On the author's estimate the avoided fee was on the order of 30 million FLR a year, more than the pool's entire service-fee revenue.

2.3 The proxy inevitability

Assume governance fixes the count. Suppose FIP.02 is amended, the Management Group chills the second identity, and a binding 5% share of active stake per entity is written into the reward script. What does a 2.4 billion FLR pool do?

It delegates the excess to independent validators, and it negotiates. The FIP.16 floor guarantees those validators 20% of rewards; a side agreement returns part of it. The P-chain sees distinct NodeIDs. The registry sees distinct entities. The reward script sees a diversified delegation. The capital is as concentrated as before, and its concentration is now invisible. Nothing in this is illegal, and nothing in it is what FIP.05 or FIP.16 meant.

This is not a hypothetical about Flare. It is the observed equilibrium of every proof-of-stake network that left liquid staking to the market: Lido reached roughly a third of Ethereum's stake through exactly this structure, a pool that owns no validators and selects all of them. A cap on the entity does not prevent it. A cap on the pool would, but the pool is a contract the Management Group cannot reach.

2.4 The FIP.16 precedent, stated precisely

The analogy that carries this document needs to be exact, because a loose version of it is wrong.

FIP.16 did not "ban MEV". It observed that MEV is a rational response to a protocol that leaves transaction ordering to whoever proposes the block, concluded that no rule against extractors would hold, and changed the protocol so that ordering is performed by a designated builder under a published mandate, with proceeds routed to FIRE. The extraction was internalized, not prohibited. The builder is initially the Foundation; Stage 2 moves the builder into Flare Confidential Compute; Stage 3 makes builder and proposer one protocol role.

The parallel holds for selection and breaks for settlement, and the document should not let the reader blur them.

	MEV (FIP.16)	Liquid staking (this RFP)
Extraction	Transaction ordering by whoever proposes	Validator selection and custody by whoever holds deposits
Why rules fail	Extractors are rational, ordering is invisible	Proxies are rational, side-deals are invisible
Protocol answer	Designated builder under a mandate	Designated router under a formula
What it needed	Consensus change to block validation	For the router: nothing at consensus level. For custody: a P-chain primitive that does not exist
Where Stage 3 helps	Directly; it is the Stage 3 deliverable	Not directly. Stage 3 changes builder and proposer roles and says stake "will continue to count towards the vote power"; it reopens staking mechanics, it does not specify contract-originated delegation

Validator selection by a private pool is rational under a protocol that leaves delegation to whoever holds the deposits, and no rule against proxy delegation will hold; that half of the analogy is exact. The custody half is a request, not a precedent.

2.5 The do-nothing baseline

Suppose governance adopts the percentage cap and the FIP.02 sanction and nothing else. Twelve months out, the arithmetic is the one Sceptre already did. The pool is 2.384 billion; 1.2 billion may sit under one entity; 1.18 billion must be delegated elsewhere. At the current reward rate on the delegated portion (Appendix B), that 1.18 billion earns on the order of 75 million FLR a year for the pool and 19 million of it goes to third-party entities as the 20% floor. A side agreement that returns half of that is worth about 9 million FLR a year to the pool and costs the entity nothing it would otherwise have had. There is no rule against it, no chain data that shows it, and no reason a rational operator would not do it. The baseline is not "the cap holds"; it is "the cap holds on paper and 1.2 billion is delegated on terms nobody can see". Any proposal in this document has to beat that baseline, not a strawman in which capital sits still.

3. Precedents and Their Verdicts

Three networks have faced this and answered differently. The answers are the design space.

Polkadot: nomination pools (2022). Protocol-native staking pools, in runtime, with no liquid token. Anyone may create a pool; members pool stake under one nominator account. It solved the participation problem (minimum 1 DOT) and did nothing about concentration, because pool operators still choose validators. Lesson: enshrining the pool without enshrining selection changes custody, not concentration.

Cosmos Hub: Liquid Staking Module (2023). Not an enshrined token but an enshrined constraint on private ones. Liquid staking providers may hold at most 25% of all staked ATOM (governance-adjustable), and every validator must self-bond at 1:250 against the liquid stake delegated to it. Private LSTs remain; the protocol caps their aggregate and forces validator skin in the game. Lesson: a protocol can bound pools without owning them, if the chain can distinguish pooled delegation from direct delegation.

Ethereum: the road not taken (2023-2025). Enshrined LSTs were argued in earnest by Ethereum Foundation researchers, first as a two-tier staking model, then as rainbow staking (heavy and light services, protocol-enforced self-limits for pools). None has shipped. The objections are the ones this document must answer: a protocol that selects validators has moved the political question into protocol parameters; a protocol token competes with private ones but does not remove them; and enshrinement adds consensus-level surface for a problem that might be solved by capping issuance. Lesson: enshrinement is a minority path, and its advocates lost the argument on the largest network. Flare's case has to be made on Flare's specifics, not on Ethereum's.

The specifics are favourable in two ways. Flare's validator set is small (179 active), and its validators are already FTSO data providers whose performance the protocol measures every reward epoch on-chain. The information an algorithmic router needs is already being computed. And Flare has already crossed the line Ethereum would not: FIP.16 designated a builder. A network that has enshrined block building has fewer principled objections left to enshrining delegation.

4. Scope Options

The sparring that produced this document started from "the protocol issues eFLR". That is one of three scopes, and the cheapest one is not it.

Scope A — Enshrined router, private tokens. The protocol publishes a delegation allocation every reward epoch: a target set of entities and weights computed from the formula in Section 6, with per-entity caps and capacity. Private LSTs continue to exist, issue their own tokens and hold their own keys. Above a threshold share of network stake, a pool's delegation is reward-eligible only if its observed allocation is within a tolerance ε of the published target. That is the enforcement point, and it has to be named, because a router that pools "must use" is a forum mandate unless one of three edges exists: the P-chain accepts delegation only from allow-listed originators, the reward script pays pool delegation only when it matches the router, or the entity registry binds pool contracts. The second is the only one available without a consensus change and without custody, and it is what "Scope A" means from here: a reward filter, not an appeal. A pool may still export, self-bond and side-deal; it then forfeits staking and FSP rewards on the non-conforming portion, which is the one lever the network already holds over every delegator on it.

The threshold is a political fact before it is a number. Sceptre is above any threshold this RFP would propose. Either the document says the largest pool is bound on day one, or the threshold is a schedule. The proposal is a schedule with a sunset: 11% at activation (binding nobody), stepping to 8% and then 5% at 25-epoch intervals, about three months each, so that the pool that exists today has two quarters to bring its allocation inside the target before its rewards depend on it. A schedule is a concession; it is also the difference between a rule Sceptre can comply with and one it has to fight.

Scope B — Enshrined token, protocol custody. The protocol issues eFLR against deposits, holds the capital under protocol keys, and delegates by formula. Private tokens compete on the market. Solves custody and selection for the capital that migrates; does nothing for capital that stays in private pools unless Scope A is also present.

Scope C — Both. The protocol issues eFLR and requires every pool above the threshold to route through the same router eFLR uses. Private pools become front-ends on protocol rails; their remaining product is distribution, integration and yield strategy on top of the base rate.

The recommendation of this RFP has changed between drafts and the change should be visible. Draft 1 recommended C staged, with B hung on the Stage 3 consensus change. Draft 3 recommends A as the product, in four deliverables that do not depend on each other: a percentage cap per entity in FIP.02/FIP.05; an on-chain allocation oracle that publishes T, Y, C_max and capacity per entity, published before it binds anything; Scope A as a reward-eligibility rule with the threshold schedule above, tested on Songbird for eight to twelve epochs; and contract-originated P-chain delegation as its own FIP, independent of the builder roadmap. B comes after that FIP, or after a reviewed FCC key set with a published image hash and a 2-of-3 that is not Foundation-only, and if neither exists it does not come. A alone leaves the four-keys problem with the pools, bounded by the contract terms the earlier dispatch asked for. That is a smaller claim than Draft 1 made, and the only one the network can ship without new custody.

4.1 Decision matrix

A recommendation without the criteria behind it is an opinion. The criteria, as this RFP weighs them:

Criterion	Scope A (router)	Scope B (token)	Scope C (both)
Bounds the pool that exists today	Yes, at the routing layer	No	Yes
Removes operator custody risk	No	For migrated capital	For migrated capital; private pools keep their keys
Consensus change required	No	Yes (§5.2 ii) or FCC custody (§5.2 i)	Same as B

Criterion	Scope A (router)	Scope B (token)	Scope C (both)
Governance surface added	Router parameters	Router parameters + custody envelope	Same as B
Time to Songbird	One to two quarters	Tied to a delegation FIP or a reviewed FCC key set	A first, B behind either
Political cost	Private pools lose selection, keep fees	Foundation competes with sFLR	Both
Regulatory exposure (\$11.10)	Low: no new asset	Higher: protocol-issued yield-bearing asset	Same as B

Two decision conditions, stated as proposals with dates the Foundation should replace with its own. If by the end of Q1 2027 no contract-originated delegation FIP is in draft, ship A on Songbird and treat B as closed until one is. If such a FIP is accepted and running on Coston2 by Q3 2027, decide C and sequence B behind the Songbird test plan in Appendix D. What this RFP cannot supply is the cost side: engineering effort for the primitive, FCC enclave provisioning, audit budget. Those numbers belong to the core team and are requested in Section 10.

5. The Settlement Problem

This is the section a reviewer from the core team will read first, and it is the one where the sparring draft was silent.

5.1 Two chains, two delegations

Flare inherits Avalanche's architecture: a P-chain that holds validator stake and delegation, and a C-chain that runs the EVM. Staking rewards for P-chain stake are computed off-chain by a public script and distributed by C-chain contracts through a mirror (PChainStakeMirror). FTSO delegation is a different mechanism entirely: WFLR on the C-chain, delegated by percentage to a data provider, no lock, rewards every 3.5 days. FIP.16 unified the weight both carry in the FSP protocols and set P-chain stake at 5×.

A liquid staking pool on Flare has to do both. Sceptre's contract holds deposits on the C-chain, delegates WFLR to FTSO providers, and moves a portion to the P-chain to stake against validators. The second step is the problem. A C-chain contract cannot construct or sign a P-chain transaction. Somebody has to export FLR from C to P, issue `addDelegator` or `addValidator`, and later import it back. That somebody holds keys. In Sceptre's case, four accounts with `DEPOSIT` and `WITHDRAW` roles, granted on 15 July 2026, that moved roughly 140 million FLR out of the pool into self-bonds with no timelock, no cap and no notice.

The four keys are not a Sceptre design flaw. They are the only way a C-chain pool can reach the P-chain today. Every liquid staking token on Flare has a version of them.

5.2 Three ways to close the gap

(i) Protocol-managed keys in Flare Confidential Compute. FIP.16 Stage 2 already plans to run the block builder inside FCC. The same enclave model can hold the P-chain keys of the eFLR pool: keys generated inside a TEE, never exported, executing a fixed policy (export, delegate per router output, import at maturity) against attested inputs. This is the author's earlier "Ulysses at the mast" proposal applied to staking rather than treasury. It is buildable on the FCC roadmap and it does not change consensus. Its weakness is the enclave image: whoever can replace it controls the pool, so the attestation and upgrade path for the image is the actual custody question.

(ii) A P-chain primitive for contract-originated delegation. The P-chain recognises a delegation whose owner is a C-chain contract address, with the mirror running in reverse: the contract emits intent, the P-chain executes it, the mirror proves execution. This is the clean answer, and it is a consensus change that nobody has scheduled. Draft 1 read FIP.16's Stage 3 sentence, that "the technical implementation of how exactly staking is handled, where the funds reside, and concrete staking processes will change", as a window for this. That reading was too generous. Stage 3 changes who builds and proposes blocks; it says stake keeps counting as vote power and validators stay the security layer; it reopens staking mechanics without saying anything about a contract as delegator. The primitive is a separate FIP with its own scope, and it should be written that way so that it is not hostage to the builder roadmap's timing, and so that the builder roadmap is not asked to carry a change it did not propose.

Mirror limit. One further constraint neither route escapes. The C-chain mirror recognises stake to at most three validators per P-chain address for rewards and vote power. A protocol pool delegating across a hundred entities therefore cannot sit on one P-chain address; it needs a vault architecture of many addresses, each holding stake to at most three nodes, with reward accounting that aggregates across them. That is engineering, not research, but it sizes the work, it multiplies the key material under (i), and it is why "one enclave, one key" in Appendix A.2 is a simplification the core team would have to correct.

(iii) Restrict eFLR to C-chain delegation. A pool that only delegates WFLR to FTSO providers needs no P-chain keys at all. It also earns at $1\times$ weight against $5\times$ for P-chain stake, and it does nothing for validator concentration, because it never touches validators. This is not a liquid staking token; it is a delegation wrapper, and it is mentioned so that it can be ruled out.

The RFP asks the core team one question above all others: will (ii) be scoped as its own FIP, and what would it cost? Everything in Section 7 assumes that either (i) or (ii) exists; Scope A assumes neither.

Between (i) and (ii) the RFP takes a position rather than leaving it as a menu. The enclave route has a single point of failure in the image and its attestation chain, and a compromise there is a loss of the pool with no recovery path except a hard fork

(§11.12). The primitive route has consensus risk, which is bounded by Songbird testing and by the fact that the network has shipped consensus changes through that path before, most recently the FIP.16 fork. A consensus change tested on Songbird is a risk the network has taken before; an enclave holding a multi-billion FLR pool is not. The primitive is the target; the enclave is acceptable only as a bridge with a published sunset. Appendix A sets out the minimum each option would have to specify for the core team to answer with more than "possible".

5.3 Liquidity: the exit queue

P-chain delegation locks for a chosen period, minimum two weeks, and cannot be withdrawn early. A liquid token over locked capital needs either an instant-redemption reserve (Sceptre's model, and the reserve the four keys drew down) or an exit queue with a redemption delay, or both. The sparring draft assumed instant, zero-fee redemption during a governance timelock; that is not achievable against locked P-chain stake without a reserve large enough to honour it.

The proposal: a laddered delegation schedule (delegations staggered so that a fixed share matures every epoch), an exit queue served in order from maturing delegations, and an instant-redemption reserve whose minimum size is a contract invariant, not an administrator's discretion. The ragequit right in Section 8 is the right to enter the queue at the head, not the right to instant exit, and its value depends on the ladder: priority in a queue whose next maturity is 180 days out is cosmetic, which is why the ladder length is a parameter with a ceiling, not a free choice.

Three things the sketch has to say to be a design. First, a discount policy. Under stress eFLR trades below its exchange rate on secondary markets, and the question is who may close that discount. Nobody with administrator access to the reserve; the invariant exists to make that impossible. External market makers, yes, that is what a discount is for. FIRE, no: a protocol buying its own pool's token to defend a peg is the protocol becoming a market maker in its own liability, and it is excluded in writing. Second, a griefing model. An attacker can inflate the queue, drain the reserve to its floor, push the secondary discount, and then present governance with a demand for a "reserve patch". The defence is that the invariant is in the immutable core and the emergency powers in Section 8 cannot reach it; the cost of the attack is the attacker's own exit at a discount; and the queue serves in order, so the attacker exits last of its own tranche. Third, numbers. Appendix C runs 5%, 10% and 20% outflows over one and thirty days against 90-, 180- and 365-day ladders with a 5% reserve. The short version: a 5% reserve absorbs a 5% one-day run outright, a 10% run clears in five to twenty days depending on the ladder, a 20% run in fifteen to fifty-eight. A ladder longer than 180 days turns every stress event into a two-month queue, and the reserve size and ladder length should be set from that table, not from the number 5.

Sceptre's buffer was advertised at about 50 million FLR and was the thing the four keys drew from. eFLR's reserve has one job beyond liquidity: to demonstrate that "invariant" does not mean "there is another function the administrator can call".

6. Architecture

6.1 Deposit and token

Users deposit FLR (or WFLR) and receive eFLR. Two accounting models exist. A rebasing token (balance grows) is simpler to reason about and breaks most DeFi integrations. A value-accruing token (balance fixed, exchange rate to FLR rises, sFLR's model) is what every lending market, AMM and yield-stripping protocol on Flare already integrates. eFLR should be value-accruing. This is not a small choice: it determines whether Spectra, Kinetic and Enosys can list eFLR on day one by cloning their sFLR configuration.

6.2 Capital sources

Two inflows. User deposits, which are the point. And, as an option the RFP puts on the table rather than assumes, protocol-owned capital: FIRE's mandate already ranks "rewards to FLR validators and stakers" as its second priority, and FIRE's captured FLR could be deposited into eFLR rather than distributed. The effect would be to seed the pool and to give FIRE a yield-bearing FLR position. Two cautions. FIRE's assets are mixed (stablecoins, FXRP, FLR), so only the FLR leg qualifies. And a pool part-owned by the Foundation's revenue entity is a pool whose neutrality will be questioned; Section 8 addresses whether the governance envelope can carry that.

The "deflationary flywheel" claim from the sparring draft is dropped. Locking FLR in a pool changes float, not supply. The fee in Section 7.2 burns a fraction of yield that is small against 3% inflation. eFLR is a supply sink in the FIP.16 sense (excluded from the inflatable base while locked) and nothing more, and the document should not say more.

6.3 The delegation router

The router is the mechanism that replaces Sceptre's validator analysts. Every reward epoch (3.5 days) it computes a target allocation of the pool's stake across eligible entities and issues the delegation instructions needed to move toward it, subject to lock schedules.

Target allocation per entity i :

$$A_i = \min(W \cdot (T_i \cdot Y_i) / \sum_j (T_j \cdot Y_j) , C_{\max,i} , K_i)$$

where W is the pool's delegable stake, T is a trust weight, Y is a net-yield weight, $C_{\max,i}$ is the per-entity cap from 6.7, and K_i is the entity's remaining delegation capacity under FIP.05's $15\times$ factor and the 300 million node maximum. Surplus above any entity's cap or capacity redistributes to the next by the same weights. Without K the router issues instructions the P-chain rejects: free delegation space

exists in aggregate (about 8.6 billion on 5 September 2026) and individual entities are full.

The formula is a target, and a target is not an instruction. P-chain delegation locks for two weeks to a year, so the pool cannot reach its target every 3.5 days. Three constraints turn the formula into a schedule. A rebalance budget: in any epoch, the router may move only capital that matured that epoch plus reserve above the invariant, and nothing else. A maximum step: no entity's allocation moves by more than a fixed share of the pool per epoch, so that a change in T or Y produces a drift, not a jump. A tracking-error bound: the distance between actual and target allocation is published every epoch, and if it exceeds a threshold for more than a set number of epochs, that is a monitoring event under Section 11, not a reason to break the lock schedule. The on-chain state is the delegation ladder; the formula is what the ladder is steered toward. Delegation is per entity, summed over that entity's registered nodes, because the entity is what the registry knows and the node is what the P-chain knows; the router must reconcile both.

Three terms, used consistently from here:

Term	Definition	Known to	Role in the router
Node	One validator with its own NodeID and stake	P-chain	Unit of delegation execution
Entity	Registered identity, one to four nodes	C-chain registry	Unit of scoring (T, Y) and of the cap C_max
Operator	The organisation behind one or more entities	Nobody on-chain	Target of the correlation rule (6.8)

The router scores and caps entities, executes against nodes, and applies the correlation rule at entity level: an entity that spreads four nodes across four hosts does not thereby escape a penalty it shares with another entity on any one of them.

6.4 Eligibility (hard filters)

An entity is eligible if, over the evaluation window, it has met the FIP.05 uptime requirement (80%; the RFP proposes the router use a stricter 95%), has been rewarded by the FSP for FTSO data provision in every epoch of the window (which is what "good enough prices" means in FIP.05 terms and is already computed on-chain), holds no active chill under FIP.02, and has a self-bond meeting the P-chain minimum. There is no slashing on Flare's P-chain; "slashing history" in the sparring spec is replaced by chill history, which is the network's actual sanction record.

6.5 Trust weight T_i

New entities enter at $T = 0.1$ and rise linearly to $T = 1.0$ over 50 reward epochs (about six months) of continuous eligibility, and for those 50 epochs their cap is one fifth of C_max. Draft 2 had T start at 0.5; at the pool sizes this document contemplates, half weight after one eligible epoch is a large delegation to a party the network has watched for four days. The purpose is to make Sybil entry expensive in

time, since a 1 million FLR self-bond does not make it expensive in capital. A window of 30 epochs (about 105 days) governs penalties: each epoch of ineligibility inside the window deducts 0.2 from T, floored at 0; eligibility restores T by 0.05 per clean epoch. An entity that goes dark for one epoch loses a fifth of its weight for four epochs and is back at full weight within a month. An entity that is chilled loses eligibility for the chill's duration and re-enters at $T = 0.5$. No lifetime exclusions; the router is not a court.

The obvious attack is identity splitting: an operator at $C_{\max}$ registers a second entity, waits out the ramp, and after six months collects delegation under two names. Two things bound it, and neither is the ramp. FIP.05's delegation factor caps total stake on a node at $15\times$ its self-bond, so a fresh entity with the 1 million FLR minimum can receive at most 14 million in delegation from anyone, router included; scaling the attack means scaling self-bond, which is capital the operator has to lock. And the cap in 6.7 is computed on the entity's share of network stake, so a second entity that shares infrastructure with the first shares its cap under 6.8. A reviewer proposed tying the ramp to self-bond size directly (T rising more slowly below a 5 million FLR bond). The RFP does not adopt it: the delegation factor already prices Sybil entry in capital, and a bond-weighted ramp would slow exactly the small honest operators the baseline exists for. What the ramp buys is time, and time is the one input a well-funded attacker cannot buy.

6.6 Net-yield weight Y_i

Flare's validators are data providers, and their reward rate varies with data quality. A router that ignores that pays bad data providers to exist. Y_i is the entity's realised net reward rate to delegators over the trailing 10 epochs, divided by the network median:

$$Y_i = (\text{gross reward rate}_i \times (1 - \text{fee}_i)) / \text{median}_j(\text{gross reward rate}_j \times (1 - \text{fee}_j))$$

Fee is the entity's declared fee, floored at 20% by FIP.16. An entity charging 35% competes against one charging 20% on net yield and loses delegation to it; an entity whose FTSO submissions earn more competes on gross and wins. Y is clamped to [0.5, 1.5] and computed on a trailing ten-epoch mean, so that one exceptional epoch moves an entity's weight by at most a tenth of its excess and the router remains a baseline, not a momentum trader. Yield-hopping, where delegation chases last epoch's winner, is a failure mode of every reactive allocator, and the window and the clamp are the two instruments against it; if either proves too loose on Songbird, both are parameters.

Y has a second problem the clamp does not solve. Once the pool is large, its own allocation is a determinant of the yields it measures: delegation raises an entity's signing weight, which raises its FSP rewards, which raises its Y, which raises its delegation. The clamp bounds the speed of that loop, not its existence. Two mitigations, both partial: Y is computed on reward rate per unit of stake, not on reward volume, which removes the first-order effect; and the cap in 6.7 bounds how

far any one entity can ride the loop. The residual is a real limit of a reactive allocator at scale, and it is one reason the router should publish its allocation for several epochs before it binds anything.

This is the part of the design that answers the objection that an enshrined pool socialises the validator set. It does the opposite. Today delegation from the largest pool follows a relationship with the pool operator, and the terms of that relationship are not public. Under the router, delegation follows three numbers every entity can see and move: uptime, data quality, and fee. An entity that improves its FTSO feeds gains delegation next epoch from a delegator that cannot be lobbied. The competition is not dampened; it is made legible and open to every entity that meets the filters, including the ones that have no business-development function at all.

6.7 Cap C_max

The cap is the point of the whole design and the parameter most likely to be fought over. The sparring draft proposed 3% of the pool per node. That is the wrong denominator. A pool that is 40% of the network with a 3% per-node cap still hands 1.2% of the network to a node in one instruction; and a cap on the pool's share says nothing about the entity's total share once its other delegators are counted.

The proposal: C_max is expressed as a share of total active network stake, per entity, inclusive of the entity's stake from all sources as the reward script already sees it, and set so that the router never pushes an entity across the binding percentage cap this RFP assumes governance will adopt in FIP.02/FIP.05. If that cap is 5%, the router's C_max is 5% minus the entity's non-router stake. The router then does with code what the Management Group cannot do with a forum: it makes the largest delegator on the network incapable of concentrating.

6.8 Correlation

Eight of Sceptre's nodes sat in one /24 on one host. A cap per entity does not see that two entities share a rack. The right rule is that entities whose nodes share an autonomous system, hosting provider or /24 with another eligible entity share one cap between them for the shared portion. The right rule is not yet a consensus input. Host and AS attribution is off-chain data (Catenalytica publishes it), it is contestable, and a router that reads it is a router whose allocation can be moved by whoever edits a hosting record. Until an oracle-grade source exists, the correlation rule is advisory: computed and published alongside the allocation, reported quarterly under Section 11, and not part of C_max. Draft 2 had it inside the cap; that was the wrong place for data the chain cannot verify.

6.9 Two yields, two routers

A Flare liquid staking pool runs two machines, and this document has so far specified one. P-chain stake earns validator rewards and carries 5× signing weight; it is locked. WFLR on the C-chain earns FTSO and FSP rewards at 1× and is liquid. A pool decides how much of its capital sits in each, and Sceptre decides it today by

discretion. The RFP proposes: the split is a parameter, initially set so that the liquid leg covers the reserve invariant plus the next epoch's expected redemptions and everything else is exported to the P-chain, because $5\times$ against $1\times$ is not a close call for yield or for security. The C-chain leg is delegated by the same entity weights as the P-chain leg, so that the router does not run two contradictory selections; a separate FTSO-only formula is possible and not proposed. Rewards on both legs are claimed by the contract each epoch and compounded into the exchange rate, with no claim role held by any party.

The consequence has to be said as plainly as 11.1 says the pool problem: if eFLR reaches 30–50% of stake, the router is not only the largest delegator but the largest selector of FTSO data providers on the network, and its entity weights decide a large share of which prices the oracle publishes. That is more power than any pool operator has held, and it is the strongest reason the allocation oracle should run in public, non-binding, for several epochs before the reward filter is switched on.

7. Economics

7.1 The base rate

The pool's gross yield is the stake-weighted average of what its delegated entities earn, less their fees (20% floor), less the protocol fee below. At today's parameters an sFLR holder sees roughly 7–8% gross on the pool, of which Sceptre takes 10% and the network's 3% inflation absorbs most of the rest in real terms. The author's estimate of Sceptre's numbers: about 228 million FLR a year of pool rewards on 2.384 billion, about 23 million of which is service fee.

eFLR's advantage over that is not the fee, and the arithmetic should be on the page. Call the gross reward rate on delegated stake g . The router pays the 20% floor on everything, leaving $0.80g$, and a 5% protocol fee on that leaves $0.76g$ to the depositor. A pool that self-validates a third of its capital (Sceptre's proportion) and passes the self-bond rewards through earns g on that third and $0.80g$ on the rest, $0.87g$ before its own 10% fee, $0.78g$ after. Under the same gross, eFLR pays the depositor about two and a half percent less than a self-validating private pool, and more if the private pool keeps the self-bond rewards for its operator instead. That gap is the price of the router paying every entity the floor and of the pool being incapable of self-bonding. eFLR does not win on yield; it wins on not being the counterparty that can take the reserve, and Section 10.7 says what follows for migration.

7.2 Protocol fee

A fee on pool rewards, proposed at 5% for discussion; Sceptre charges 10%, typical LST fees run 10–20%, and the right number is whatever covers the reserve and leaves eFLR's net rate above the private pools' without a self-bond subsidy. Two

destinations, and the sparring draft's 50/50 split is replaced with something the numbers can carry.

The first destination is the instant-redemption reserve, until it reaches its invariant size (proposed: 5% of pool). Once the reserve is full, the fee goes to the second destination, and here the RFP has to name a conflict it created. FIRE is administered by the Foundation. A fee from eFLR to FIRE is, in practice, a fee to the Foundation, and it gives the Foundation a direct financial stake in eFLR's growth. Every parameter in Section 6 that makes eFLR more attractive to depositors (a looser uptime filter, a softer Y clamp, a higher cap) raises that fee. If the same Foundation also holds the custody keys, the party that sets the router's neutrality is paid for the router's size. That is not a hypothetical; it is the structure.

The fee's destination is therefore conditional. If the custody condition in Section 8 is met, and no single party including the Foundation can replace the enclave image or the upgrade key, the fee may go to FIRE under its existing mandate, because FIRE's priorities two through four are the ones an eFLR fee should fund. If the custody condition is not met, the fee is burned. And in either case one combination is excluded: FIRE may not seed the pool (Section 6.2) while FIRE receives the fee. A protocol that is a depositor in its own pool and the recipient of its fees has closed a loop in which its neutrality is not a matter of keys at all. One or the other; the RFP's preference is the fee, because seeding is the one that cannot be undone. The 5% itself stays a discussion number until the reserve simulation in Appendix C has been run on Songbird data; it is not cemented here. Burn benefits every FLR holder equally and gives no party an interest in the pool's size. The RFP does not propose a separate "insurance fund". Flare's P-chain does not slash, so there is no principal loss to insure against from validator conduct. The arithmetic for the risks that remain is short: a 2 billion FLR pool at 7% gross yields about 140 million FLR a year, of which 5% is 7 million; an enclave compromise loses the pool. A fund that claims to cover a loss three hundred times its annual inflow is marketing, and "principal-protected" does not appear in this document. What a compromise does get is \$11.11.

7.3 Who earns what

The depositor earns the base rate net of 20% entity fees and 5% protocol fee, holds a token the counterparty cannot drain, and gives up the possibility of the above-market yield a self-validating pool can offer. The entity earns the 20% floor on router delegation, allocated by uptime, data quality and fee discipline rather than by relationship with a pool operator; small entities that perform get delegation they cannot market for today. The protocol earns a fee stream into FIRE and, more importantly, a validator set whose largest delegator is incapable of concentration by construction. The private pool operator loses the yield edge from self-bonding and the discretionary selection business, and keeps distribution, integrations and strategy layers.

7.4 What this does to the 20% floor

FIP.16's floor exists to make independent entities viable. A router that delegates by net yield puts every entity at the floor, because any fee above it loses delegation. That is the intended outcome and it should be said plainly: the router makes the floor the ceiling. If governance wants a market in entity fees above 20%, the Y weighting has to be softened. The RFP's position is that it should not be.

8. Governance Envelope

The sparring conversation's own objection: enshrining the pool moves the bottleneck from four private keys to whoever can change the contract. On Flare that is, today, the Foundation, through the governance timelock on every system contract. The FIRE precedent is instructive and not reassuring: FIRE is administered by the Foundation with a "negative governance" veto that requires 50% of the inflatable supply to vote, a threshold no vote has reached.

The envelope this RFP proposes has four parts, and the fourth is the one that matters.

Immutable core. Deposit, redemption, exit queue, the reserve invariant, and the cap formula are non-upgradable. If they are wrong, the pool is wound down and redeployed; it is not patched.

Parameter set under timelock. T ramp length, penalty size, Y clamp, uptime threshold, fee, reserve size, correlation rule: adjustable, with a 21-day on-chain timelock and public diff before effect.

Ragequit during timelock. Any pending parameter change opens a window in which eFLR holders may enter the exit queue at the head, ahead of ordinary redemptions, served from the reserve and maturing delegations. It is not instant exit; Section 5.3 explains why it cannot be. It is guaranteed priority.

Custody outside the Foundation. This is the part the sparring draft's dual-veto does not solve. A veto over parameters is worth little if the enclave image, the router's data feeds and the upgrade key are held by the same party. The four buckles from the author's FIRE essay apply verbatim: the upgrade key, the mandate governance, the enclave image, the input feeds. Each must have a documented holder, and for at least the enclave image and the upgrade key the holder must not be the Foundation alone. A 2-of-3 across Foundation, an elected entity committee (FIP.16's own joint-governance mechanism, Songbird-then-Flare election) and a time-delayed community key is the minimum. If the Foundation will not accept that, Scope A (router without protocol custody) is the honest fallback, and the four keys stay with the pools, bounded by the contract terms the earlier dispatch asked for.

Transition. A 2-of-3 that does not exist at launch is a promise, and Section 3 of the author's FIRE essay already said what promises are worth. The proposal is staged so

that each stage is a precondition for the next, not a hope after it. At launch, on Songbird, the Foundation holds the keys alone, under the 21-day timelock and the ragequit right; this is acceptable because Songbird is the test and the amounts are SGB. Before any Flare deployment, the entity committee exists: elected by the mechanism FIP.16 §4.5.1 already specifies (a Songbird vote to shortlist eight, a Flare vote to seat four) but without FIP.16's precondition of a 50%-of-supply vote to unlock it, which has never been met and was designed not to be. Before the pool exceeds a threshold share of active stake (proposed: 10%), the community key exists: a time-delayed multisig whose signers are elected by eFLR holders, with the power to block a parameter change or an image replacement and no power to initiate either. If the second or third stage is not in place when its trigger arrives, deposits close until it is. Closing deposits is the enforcement; there is no other.

Custody matrix. Section 8 is an essay until it is a table. The one below is the proposal; every cell is a decision the Foundation can accept, reject or replace, and a cell left blank at launch is a reason not to launch.

Asset or action	Who signs	Delay	If the key is lost
Router parameters (T, Y, filters, cap, ladder)	2-of-3: Foundation, entity committee, community key	21 days, epoch-boundary effect	Status quo holds; missing party is re-elected
Enclave image hash (route i)	2-of-3, with the community key required	21 days + one full ladder before old key retires	Old image continues; no new deposits until replaced
Reserve invariant, exit queue, cap formula	Nobody; immutable	—	Wind-down path below
Pause new deposits and new instructions	Any one of the three	Immediate; expires after two epochs unless renewed by 2-of-3	—
Move capital	Nobody; only the bridge, only per allocation	—	—
Wind-down	2-of-3	21 days; then deposits close, ladder runs off, redemptions at exchange rate	—

Who the parties are, concretely. The Foundation is the Foundation. The entity committee is the four seats FIP.16 §4.5.1 already describes, elected by SGB shortlist and FLR vote, seated without the 50%-of-supply precondition; its members are entities the router pays, which is a conflict, and the reason it holds a veto and not an initiative. The community key is a 7-of-12 multisig whose signers are elected annually by eFLR holders weighted by balance, generated in a public ceremony, rotated at each election, with a 14-day delay on every signature. The FCC image, if route (i) is used, is a reproducible build whose hash is published on-chain by the 2-of-3 before

attestation is accepted, with a written policy that a CVE in the TEE triggers a pause, not a patch, until a new image passes the same path. Immutable core and "wind down if wrong" is an event at 30–50% of stake, not an upgrade path, which is why the pause power exists and why wind-down is specified as a ladder run-off with redemptions at the exchange rate, not a redeploy.

Deadlock. Three parties will disagree. The rule is that the status quo wins: any change to parameters or to the enclave image needs two of three, so one party can block, and a block is not a crisis, it is the design working. Where all three want a change and differ on the value, the most conservative proposal takes effect (the lowest cap, the strictest filter, the longest timelock), and the others may propose again after one reward epoch. Where the disagreement is about an emergency (a live exploit, an image with a known bug), the only emergency action available to any single party is to pause new deposits and new delegation instructions; existing delegations mature on schedule and the exit queue keeps running. Nobody can move capital alone, in an emergency or otherwise. That is the whole point of the envelope, and an emergency power that could override it would be the four keys again.

9. Composability

eFLR is a base asset. Everything below it in the stack (fixed-term lending, options collateral, yield stripping) exists or is proposed elsewhere, and this section decides what the protocol builds and what it leaves.

The sparring draft proposed native PT/YT splitting in the protocol. This RFP recommends against it. Spectra already runs yield tokenisation on Flare against sFLR, with live pools across maturities, and the Fixed-Term Lending RFP already specifies PT-as-collateral on Spectra's rails. An enshrined splitter duplicates a live product, adds contract surface to the core, and forces the protocol to maintain a maturity calendar. What the protocol should do instead is make eFLR trivially strippable: value-accruing accounting, a clean exchange-rate oracle, no transfer hooks, no rebasing. Spectra then lists eFLR as it listed sFLR, and PT-eFLR becomes the collateral the lending RFP wants: a claim on protocol-custodied FLR with no operator key risk, which is the property that makes it institutional-grade rather than merely fixed-rate.

The author's conflict is on the table here: the recommendation favours a protocol the author uses and a market the author wants to exist. The counter-argument, that enshrined stripping removes a dependency on a third party, is real, and Section 10 keeps it open.

FAssets. The largest consumer of FLR collateral on the network is not a lending market; it is the FAssets system, where FLR sits in agents' collateral pools behind FXRP. eFLR in that role would replace an asset with operator key risk (sFLR) with one without it, which is an argument FAssets governance should hear, and it comes with a liquidity condition: an FAssets liquidation that receives eFLR has to be able to

turn it into FLR faster than the exit queue allows. Two ways to meet that. A haircut on eFLR's collateral value sized to the secondary-market discount observed on Songbird under stress, which is the honest way and needs no new mechanism. Or a liquidation lane in the reserve, ahead of ordinary redemptions and behind ragequit, capped at a share of the reserve the invariant can spare. The RFP proposes the first and leaves the second to the FAssets team, and it does not propose collateral factors; those are FAssets governance's to set against data that does not yet exist.

10. Open Questions

Ordered by how much the answer changes the design.

1. **Will the core team scope contract-originated P-chain delegation as its own FIP?** It is not on the Stage 3 path (§2.4, §5.2) and should not be made to wait for it. Until it exists, Scope B depends on FCC-managed keys and the enclave image becomes the custody question.
2. **Can the router's inputs be made oracle-grade?** Uptime and FTSO reward data are on-chain; host and AS attribution are not. The correlation rule (6.8) needs a source the protocol can verify or it stays advisory.
3. **Does the registry refuse a fifth node?** The earlier dispatch could not confirm it. If the registry enforces the count, Scope A can bind pools at the registry; if not, the mirror service is the enforcement point and phase 3 is a prerequisite.
4. **What threshold triggers mandatory routing under Scope A?** Cosmos chose 25% of all stake for the aggregate of liquid providers. Flare's largest pool is at 11%. A per-pool threshold of 5% of active stake is the number consistent with the entity cap; the RFP asks whether it should be lower.
5. **Should FIRE seed the pool?** Section 6.2 puts the option forward and names the neutrality cost. The answer depends on Section 8's custody outcome.
6. **Native stripping or Spectra rails?** Section 9 recommends rails. The dependency argument against it deserves a written answer from the Foundation, not from this author.
7. **What happens to sFLR?** Three paths, and the RFP should be plain about which ones work. A soft path (eFLR launches, sFLR stays, a conversion at the snapshot exchange rate with a small bonus for early movers) is politically costless and, on the Cosmos and Ethereum evidence, does not move capital; holders stay where their integrations are. A hard path (Scope C: sFLR must delegate through the router above the threshold) removes the concentration and leaves Sceptre its 10% service fee, its front-end and its integrations, which is why it is the path a rational Sceptre should prefer to a percentage cap it has to route around. A third path, freezing or force-converting the sFLR contract, is excluded here and should be excluded in writing by the Foundation: it would establish that a protocol can seize a private contract's deposits by vote, and that precedent is worth more to an attacker than any pool. The honest statement is that without the hard path eFLR does not reach critical mass, and the decision between soft and hard is the Foundation's political decision, not a

technical one. The hard path's arithmetic is in 7.1: a routed sFLR loses the self-bond edge (about 0.87g to 0.80g before fees) and keeps its 10% service fee, its integrations and its front-end.

8. **Songbird first?** Every consensus-touching FIP has shipped on Songbird first. The router (Scope A) can run on Songbird against SGB staking within a quarter of a decision; the token (Scope B) should not ship on Flare before it has run through at least two Songbird reward-epoch cycles with the exit queue under load.
 9. **What is a pool?** The threshold in Scope A binds "a pool above x% of active stake", and a pool is not an on-chain object. One issuer can split deposits across contracts, wrap a token in a vault, or present as a front-end over another pool. The proposal: a pool is any set of contracts whose delegated stake is controlled by the same set of privileged roles, as the registry already identifies entities by controlling address, and the burden of showing separateness lies with the operator. That is a definition the reward script can apply and a lawyer can argue with; without one the threshold is arbitrage.
 10. **What does it cost?** Engineering effort for the P-chain primitive, FCC provisioning for the enclave route, audit scope for the immutable core, and the reserve's opportunity cost. This RFP has no basis for those numbers and does not invent them. The decision matrix in 4.1 is incomplete without them, and the request is that the core team fill that column before the scope decision, not after.
 11. **Will FAssets governance accept eFLR as pool collateral, and on what haircut?** Section 9 argues it should; the number depends on Songbird stress data that does not exist yet.
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11. Failure Modes

Hardest first. Before the list, who measures: every test below carries a date and a threshold, and a Foundation that measures its own project will find it succeeding. The proposal is that the metrics (pool share of active stake, share of pooled stake routed, entity count receiving router delegation, exit-queue depth, secondary-market discount) are published as time series by a party that does not hold keys, on the pattern Flare Metrics and Catenalytica already follow, quarterly, on the forum. Three consecutive missed quarterly thresholds open a review in which the three custody parties of Section 8 must, within one reward-epoch cycle, choose one of: adjust parameters, change scope, or wind down. Under Scope C "routed" means delegation that passed through the router from any pool, sFLR included; eFLR's own share is a separate metric and a weaker one.

11.1 The enshrined pool is the largest pool. If eFLR works it becomes 30–50% of active stake, and the largest delegator on the network is a contract whose parameters are set by governance and whose keys are held by whoever holds them. Every objection to Sceptre's 11% applies at three times the size. The design's defence is that the router cannot concentrate (6.7) and the keys cannot be used

discretionarily (8). Both defences are exactly as strong as their implementation and no stronger. If the four buckles are not closed, this RFP has proposed a bigger Sceptre with a Foundation logo. Test: before mainnet, an independent review of the enclave image and key holders, published; if any single party can replace the image, do not ship Scope B.

11.2 The router becomes the politics. Every parameter in Section 6 is a decision about who gets delegation. Entities will lobby for the uptime threshold, the Y clamp, the correlation rule. The forum fight about Sceptre's identities becomes a permanent fight about T and Y. Mitigation: parameters under timelock with ragequit, so that a lost fight is an exit; and a rule that parameter changes take effect only at reward-epoch boundaries with a 21-day notice. This does not remove the politics. It gives it a schedule.

11.3 The proxy problem survives. A cap per entity inside the router does not stop a cartel of twenty entities from taking 60% of it. Section 6.8's correlation rule catches shared infrastructure; it does not catch shared ownership on separate infrastructure. The document should not claim otherwise. The honest statement: enshrinement removes the pool operator as the concentrating actor and leaves validator-level collusion where every proof-of-stake network leaves it, at the limit of what code can see. An identity-staking rule, in which an entity bonds capital against its declared independence and loses it when a later-established breach shows otherwise, is the next instrument. Its buildability on Flare's P-chain is unresolved and it is not in this RFP; it is announced here as the follow-up so that C_max plus a correlation report is not mistaken for a solution to a problem it only halves.

11.4 Nobody migrates. sFLR is integrated in every venue on Flare, has a liquid market, and can offer above-router yield by self-validating. eFLR launches with a lower yield and no integrations. Capital does not move for safety it has not been made to feel. Under Scope B alone this is likely; the Cosmos LSM's 25% cap was never binding because private ATOM liquid staking never reached it. Under Scope C the question is moot for concentration (the pool routes through the protocol either way) and open for custody. Test: eighteen months after launch, eFLR below 10% of liquid-staked FLR under Scope B is a failed launch; under Scope C the metric is share of pooled stake routed, and below 80% means the threshold in 10.4 was set too high.

11.5 The settlement gap is not closed. If neither FCC-managed keys nor a P-chain primitive ships, Scope B is a private LST run by the Foundation, with the same four keys under a different name. In that case ship Scope A only, and say so.

11.6 Liquidity crisis. A run on eFLR during a market drawdown empties the reserve, the exit queue extends to the longest outstanding lock (up to a year), and eFLR trades at a discount on secondary markets. This is not a failure of the design; it is what a liquid token over locked stake does under stress, and every LST has done it. The failure is if the discount is used as an argument for administrator intervention in the reserve. The invariant in 5.3 exists to make that intervention impossible, and the document should expect the demand for it.

11.7 The 20% floor collapses to the floor. Section 7.4: the router makes 20% the ceiling. If the network's intention in FIP.16 was to allow entities to price above the floor, the router contradicts it. Governance should decide this on purpose.

11.8 Ethereum was right. The strongest external argument is that enshrinement was rejected on the network with the most research behind it, and that the rejection was on principle: a protocol that selects validators has taken a political role. Flare's answer is that it took that role in FIP.16 already and that its validator set is small enough to make an algorithmic router auditable. If the Foundation does not accept the first half of that answer, this RFP has no ground to stand on, and the right document is a FIP.02 amendment, not this one.

Threat model. The failure modes below are the ones the design is built around. Appendix E lists the adversaries and what each can reach; the short list is TEE compromise, poisoned host data, governance capture of Y, queue griefing, and reward-script desynchronisation between C-chain and P-chain.

11.9 The router's inputs are wrong. The router allocates on FSP reward data and uptime. If the FTSO layer is compromised, the router delegates toward whoever is gaming it, at scale, every epoch. Two limits on that. A compromised FTSO is a compromised network; the router is not the largest thing that breaks, and a document that solved it here would be overclaiming. And the router can be made to fail closed: if any entity's inputs move more than a bound in one epoch, or the network median moves more than a bound, the router issues no new instructions that epoch and existing delegations mature on schedule. A router that pauses is a delegator that stops, which is what Sceptre's four keys could not be made to do. Cross-checking against a second data source is desirable and, for host attribution, not yet oracle-grade (Open Question 2); the fail-closed rule does not depend on it.

11.10 Regulators read "protocol-issued yield-bearing token" and see a fund. eFLR is a claim on a pool managed by a published algorithm with a fee. Under MiCA and under most securities analyses that description is close enough to a collective investment scheme to require an answer, and the answer "there is no issuer" is weaker for a Foundation-deployed contract than for a private one. Scope A carries none of this; it issues nothing. The RFP's position is that a written legal analysis for the EU, UK and US precedes any Scope B deployment on Flare, and that its absence is a reason to ship A alone, not a reason to ship B and hope.

11.11 The token is the Foundation's. A protocol-owned LST with the Foundation in the custody chain is a different object from sFLR in one more way than the regulatory one: its brand is the network's. A depeg, a queue, a compromised image are not a startup's failure, they are Flare's, on the network's own name, with the network's own treasury adjacent. sFLR can fail and Flare survives it. eFLR cannot fail without the network wearing it. That is a reason to ship A, whose failure is a rule that did not bind, before B, whose failure is a token that did not hold.

11.12 The enclave is broken. Under §5.2 (i) the keys live in an FCC image. If the image is compromised and the keys extracted, the P-chain delegations can be

redirected as they mature and the reserve can be drained. There is no fund that covers this (§7.2), the loss falls on eFLR holders, and recovery is a hard fork or nothing. This is the argument for the primitive over the enclave, stated as a failure mode so that it is not lost: a route whose failure is total is a bridge, not a destination. Under (ii) the equivalent failure is a consensus bug in the primitive, which is why B waits for two Songbird cycles.

12. What Ships First

A percentage cap per entity in FIP.02/FIP.05, which needs no part of this document. Then the allocation oracle on Songbird, publishing and binding nothing, so the network can see what the formula would do before it does it. Then Scope A as a reward-eligibility rule, threshold on the schedule in Section 4, eight to twelve Songbird epochs, then Flare. In parallel and independently, a FIP for contract-originated P-chain delegation. Scope B only behind that FIP or a reviewed FCC key set with the custody matrix in Section 8 filled in; and if neither, an explicit decision that B is not coming, which is a legitimate outcome of this RFP. The order matters because A bounds the pool that exists today with a lever the network already holds, and B does not exist until somebody builds a settlement primitive that nobody has yet proposed.

The one thing this RFP asks the network not to do is to treat the Sceptre case as closed when the second identity is deregistered on 16 September. The identity was never the problem. The 2.4 billion is still there, still uncapped, and the next pool to reach the ceiling will do the arithmetic Sceptre did.

Appendix A — Minimum Specification for the Settlement Bridge

Written from outside the codebase, as the questions a core reviewer would have to answer for each route. Where a mechanism is named, it is the Avalanche-derived mechanism Flare's go-flare inherits; where it is not, the RFP does not know.

A.1 P-chain primitive (§5.2 ii). A new P-chain transaction type, or an extension of `AddPermissionlessDelegatorTx`, whose reward owner and change owner are a C-chain contract address rather than a P-chain key, and which the P-chain accepts only when accompanied by a proof that the named contract emitted a matching intent (entity, nodes, amount, duration) in a finalised C-chain block. The proof can follow the pattern `PChainStakeMirror` already uses in the other direction. Intents are emitted only at reward-epoch boundaries, batched per epoch by the router, so the transaction count is bounded by the entity count, not by depositor activity; a contract cannot flood the P-chain because the contract only speaks once per epoch. C-chain reorgs are not a concern past finality, and the P-chain acts only on finalised intents. At

maturity the P-chain returns principal and rewards to the contract's atomic-memory address, and the router imports it in the next epoch. What the core team would need to state: whether the atomic export/import path can be driven without a P-chain signer at all, whether the mirror's verifier set is the right trust base for the reverse proof, and what the Stage 3 consensus changes do to any of this.

A.2 FCC enclave (§5.2 i). A TEE image that holds the pool's P-chain key material (many addresses, three validators each, per the mirror limit in §5.2), generated inside the enclave, with remote attestation of the image hash published on-chain and checked by the eFLR contract before any instruction is honoured. The image executes one policy: read the router's target allocation from a finalised C-chain block, construct the export, delegate and import transactions that move toward it, and sign nothing else. The RFP would need the Foundation to state which TEE (the FCC roadmap will decide this, and the answer determines the side-channel history the design inherits), who can publish a new image hash and under what delay, and what happens to in-flight delegations when an image is retired: the honest answer is that they mature under the old key and the new image must be able to import them, which means the key has to be transferable between images or the retirement has to wait a full lock ladder. That constraint alone is an argument for A.1.

A.3 Either route. The router's target allocation is the only input the bridge accepts, the bridge's only outputs are delegations and imports, and both are visible on-chain every epoch. Anyone can recompute the allocation from public inputs and compare. A bridge whose behaviour cannot be recomputed by an outsider is a key, whatever it is called.

Appendix B — Numbers, with dates

All figures below are the ones the argument leans on. Estimates are marked and their derivation is in "The Ceiling on Owning" (6 September 2026); this appendix does not recompute them.

Figure	Value	Source, date
Active stake	21.51bn FLR	Flare Metrics, 5 Sep 2026
Active validators	179	Flare Metrics, 5 Sep 2026
Free delegation space	8.6bn FLR	Flare Metrics, 5 Sep 2026
Largest entity (Bifrost)	1.14bn FLR, four nodes under the 300M node cap	Flare Metrics, 5 Sep 2026
Sceptre pool	2.384bn FLR, 11.08% of active stake	Ceiling on Owning, 6 Sep 2026
Entity ceiling	1.2bn FLR (4 × 300M)	FIP.05, FIP.16 fork 14 Jul 2026
Sceptre delegated share	66.4% of pool	Ceiling on Owning (estimate)

Figure	Value	Source, date
Pool rewards, annualised	~228M FLR/yr	Ceiling on Owning (estimate, epochs 426-428)
Rewards on delegated portion	~151M FLR/yr	Ceiling on Owning (estimate)
20% fee on delegated portion	~30M FLR/yr	Ceiling on Owning (estimate)
Sceptre service fee (10%)	~23M FLR/yr	Ceiling on Owning (estimate)
Sceptre advertised buffer	~50M FLR	Sceptre app, pre-July 2026
Capital moved to self-bonds	~140M FLR, seven tranches	Forum proposal §1.3, 31 Aug 2026
Gross pool yield "today"	7-8%	Sceptre UI, September 2026; varies by page and date, treat as a range
Inflation	3%	FIP.16
Do-nothing baseline (\$2.5)	$1.18\text{bn excess} \times \sim 6.3\% \times 20\% \approx 15\text{--}19\text{M FLR/yr}$ in third-party fees	derived from the above
eFLR vs self-validating pool (\$7.1)	0.76g vs 0.78g net to depositor	derived

Bifrost is the live demonstration of the document's first claim: the node ceiling already touches entity concentration (four nodes at cap is 1.14bn) and touches the pool not at all.

Appendix C — Liquidity under stress

A pool normalised to 1.0, a reserve of 5%, delegated capital laddered uniformly over L days so that $1/L$ of it matures daily, no re-delegation while the queue is non-empty, redemptions served from the reserve first and maturities second. Outflow arrives in one day or spread over thirty. The table gives days until the queue is empty.

Ladder L	Outflow 5% / 1 day	5% / 30 days	10% / 1 day	10% / 30 days	20% / 1 day	20% / 30 days
90 days	1	30	5	30	15	30
180 days	1	30	10	30	29	36
365 days	1	30	20	35	58	65

Readings. A 5% reserve absorbs a 5% one-day run and nothing larger. A 90-day ladder clears a 20% run in two weeks; a 365-day ladder takes two months, during which the token trades at whatever discount the secondary market sets, and every day of that discount is a day of pressure on the invariant. The reserve size and the ladder ceiling are the two parameters the Songbird run should set, from observed

redemption behaviour, before the Flare deployment. The model omits secondary-market absorption (which shortens the queue) and reflexive exits triggered by the discount (which lengthen it); both should be added on Songbird data.

Appendix D — Songbird test plan

"Two epochs under load" is a phrase, not a plan. Before Scope A binds on Flare: eight to twelve reward epochs on Songbird with the allocation oracle publishing and the reward filter off, then four with it on; a forced parameter change through the full timelock with the ragequit window exercised; a measured tracking error against the target every epoch, with the step and budget constraints of 6.3 visible in the ladder. Before Scope B on Flare, additionally: an artificial run of 10% and 20% of the pool against the reserve and queue, with observed secondary-market discount; an image rotation under route (i) through the custody path in Section 8, including a full ladder run-off under the old key; a mirror test with the pool's stake spread over more than three nodes per address to confirm the vault architecture accounts correctly; a pause by a single party and its expiry; and a deliberate desynchronisation between the reward script's view of the pool and the P-chain's, to see what the accounting does.

Appendix E — Threat model

Adversary	Reach	Bound by
TEE compromise (route i)	Extract keys, redirect maturing delegations, drain reserve	Nothing after the fact; only by choosing route (ii) or accepting the loss (11.12)
Poisoned host / AS data	Move correlation output	Correlation is advisory (6.8); no consensus effect
Governance capture of Y or filters	Steer delegation to a faction	2-of-3 with community veto; timelock; ragequit; step limit (6.3)
Queue griefing	Drain reserve to floor, force discount, demand a patch	Immutable invariant; in-order queue; attacker exits at own discount (5.3)
Reward-script desync C/ P	Pool paid on stale or wrong allocation	Tracking-error publication; fail-closed rule (11.9); Appendix D test
Identity splitting	Second entity, second cap	15× delegation factor; T ramp from 0.1; one-fifth cap for 50 epochs (6.5)
Validator cartel on separate infrastructure	Aggregate share across friendly entities	Not bound here; identity-staking follow-up (11.3)
Foundation as sole key holder	Everything	Custody matrix (8); if unmet, Scope A only

13. Sources

- FIP.05 — Update Services, Limits, and Rewards Required for Staking (accepted 8 Sep 2023): node/entity limits, delegation factor, uptime, "not enforced by the P-chain staking mechanism but by the mirroring service ... until phase 3". proposals.flare.network/FIP/FIP_5.html
- FIP.16 — Restructure FLR Tokenomics for Long-Term Network Sustainability (accepted 24 Apr 2026): inflation 5%→3%, FIRE, single-builder MEV roadmap (Stages 1-3), 5× P-chain signing weight, node cap 200M→300M, 20% minimum entity fee, FIRE negative governance. proposals.flare.network/FIP/FIP_16.html
- FIP.02 — FTSO Management Group (chill / second-strike ban). proposals.flare.network/FIP/FIP_2.html
- Flare, Staking Phase 2 (Nov 2023); Flare Developer Hub, FLR staking documentation (5% vote-power reward cap per validator)
- Kiln, Flare (FLR) validator documentation: no slashing on P-chain; delegation lock 60-365 days; C-chain vs P-chain mechanics
- Forum proposal, Jon-Sn0w, forum.flare.network, 31 Aug 2026 (Rotko/Sceptre multiple identities; role grants 15 Jul 2026; withdrawal→self-bond sequences)
- Sceptre, "Sceptre response and remediation status", 4 Sep 2026; Sceptre, Monthly recap May 2026 (FIP.16 impact, own-validator plans)
- Flare Metrics, 5 Sep 2026: 21.51bn active stake, 179 validators, 8.6bn free delegation space, largest entity Bifrost 1.14bn
- Janus Dispatch, "The Ceiling on Owning" (6 Sep 2026): pool 2.384bn / 11.08%, fee-avoidance estimate ~30M FLR/yr, four-keys critical path, proposed contract bounds
- Janus Dispatch, "From Fee-Burn to Sovereign Capital" (19 Jul 2026): FIRE mandate ordering, autonomous-contract-in-FCC model, the four buckles
- Cosmos Hub, Liquid Staking Module (governance-approved 2023): 25% global liquid staking cap, validator bond factor 250
- Polkadot, Nomination Pools (Nov 2022): protocol-native pools, no liquid token
- Ethereum: V. Buterin, "Should Ethereum be okay with enshrinement?" (Sep 2023); "Possible futures of the Ethereum protocol, part 3: The Scourge" (Oct 2024); B. Monnot, "Unbundling staking: towards rainbow staking" (ethresear.ch, Feb 2024)
- Avalanche validator documentation: 2% minimum delegation fee, 5× delegation factor (for comparison with Flare's 20% / 15×)

Figures marked as author's estimates are derived in "The Ceiling on Owning" and reproduced without recomputation.

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